

This thesis presents a C++ interface for two established graph generators, nauty/geng and plantri. These tools provide efficient generation of general graphs and planar graphs, but their native C interfaces are difficult to combine with modern C++ code and graph algorithms. The thesis designs and implements a unified wrapper that exposes both generators through a common, type-safe API. Generated graphs are represented by lightweight views of the internal generator data, which avoids copying and allows callback-based filtering, pruning, and output processing during generation. The interface is compatible with the Boost Graph Library, so users can apply standard graph algorithms directly to generated graphs. The resulting library makes high-performance graph generation easier to use in experimental and algorithmic graph theory software.